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DATA GENERATION DEVICE, DATA GENERATION METHOD, NON-TRANSITORY STORAGE MEDIUM STORING DATA GENERATION PROGRAM, AND TRAFFIC SERVICE PROVIDING SYSTEM

Abstract

A data generation device includes a communication device and processing circuitry. The communication device is configured to obtain, from sensors, observational datasets that have been obtained by continuously observing an object from different positions. The processing circuitry is configured to select reference data based on accuracy information related to the observational datasets, and generate a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data. The synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-030137, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.
BACKGROUND
1. Field
[0002] The present disclosure relates to a data generation device, a data generation method, and a non-transitory storage medium storing a data generation program for generating a dataset that is obtained by synchronizing observational datasets in the real world necessary for re-creation in a system that re-creates real-world traffic environments in a virtual space. The present disclosure also relates to a traffic service providing system that uses a synchronized observational dataset.
2. Description of Related Art
[0003] A digital twin is a technology that re-creates an environment identical to the real world in a virtual space. Particularly, a digital twin that re-creates real-world traffic environments in a virtual space is referred to as a traffic digital twin.
[0004] When constructing a digital twin using observational datasets, the observational datasets need to be synchronized with each other. Japanese Laid-Open Patent Publication No. 2020-160568 discloses a technology for synchronizing observational datasets obtained by capture from various viewpoints.
[0005] In the above-mentioned publication, the selection of reference data does not take data accuracy into account. If the reference data is chosen without considering its accuracy, there is a risk that the accuracy of a synchronized observational dataset may decrease depending on the accuracy of the selected reference data.
SUMMARY
[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key characteristics or essential characteristics of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.
[0007] A data generation device according to an aspect of the present disclosure includes a communication device and processing circuitry. The communication device is configured to obtain, from sensors, observational datasets that have been obtained by continuously observing an object from different positions. The processing circuitry is configured to select reference data based on accuracy information related to the observational datasets, and generate a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data. The synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.
[0008] A data generation method according to an aspect of the present disclosure includes obtaining, from sensors, observational datasets that have been obtained by continuously observing an object from different positions. The data generation method includes selecting reference data

based on accuracy information related to the observational datasets. The data generation method includes generating a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data. The synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

[0009] A non-transitory storage medium storing a data generation program according to an aspect of the present disclosure stores a data generation program configured to cause a communication device to execute obtaining, from sensors, observational datasets that have been obtained by continuously observing an object from different positions. The data generation program is configured to cause processing circuitry to execute selecting reference data based on accuracy information related to the observational datasets. The data generation program is configured to cause the processing circuitry to generate a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data. The synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

[0010] A traffic service providing system according to an aspect of the present disclosure includes a data generation device, a controller, and providing devices. The data generation device includes a communication device and processing circuitry. The communication device is configured to obtain, from sensors, observational datasets that have been obtained by continuously observing an object from different positions. The processing circuitry is configured to select reference data based on accuracy information related to the observational datasets, and generate a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data. The synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained. The data generation device is configured to send the synchronized observational dataset to the controller. The controller is configured to generate predicted moving body information using the received synchronized observational dataset. The controller and the providing devices are configured to communicate with each other, thereby providing a traffic service that assists a user of each of the providing devices based on the predicted moving body information.

[0011] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** is a schematic diagram illustrating the configuration of a traffic service providing system according to an embodiment.

[0013] FIG. **2** is a sequence diagram illustrating the communication executed by the vehicle, the controller, the data generation device, and the roadside camera in the traffic service providing system shown in FIG. **1**.

[0014] FIG. **3** is a flowchart illustrating the flow of processes executed by the data generation device in the traffic service providing system shown in FIG. **1**.

[0015] FIG. **4** is a diagram illustrating an example of obtaining observational datasets using roadside cameras of the traffic service providing system shown in FIG. **1**.

[0016] FIG. **5** is a diagram illustrating the first still image dataset obtained by the first roadside camera shown in FIG. **4**.

[0017] FIG. **6** is a schematic diagram illustrating the positions of the first vehicle and the second vehicle calculated by the data generation device from the first still image dataset shown in FIG. **5**.

[0018] FIG. **7** is a diagram illustrating the second still image dataset obtained by the second roadside camera shown in FIG. **4**.

[0019] FIG. **8** is a schematic diagram illustrating the positions of the first vehicle and the second vehicle calculated by the data generation device from the second still image dataset shown in FIG. **7**.

[0020] FIG. **9** is a schematic diagram illustrating the relationship between the video datasets before the synchronization process has been executed by the data generation device in FIG. **1**.

[0021] FIG. **10** is a schematic diagram illustrating the synchronized observational datasets synchronized by the data generation device of FIG. **1**.

[0022] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0023] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0024] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0025] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0026] An embodiment of a traffic service providing system **10** and a data generation device **100** of the traffic service providing system **10** will now be described with reference to FIGS. **1** to **10**.

Overview of Traffic Service Providing System **10**

[0027] Referring to FIG. **1**, the traffic service providing system **10** includes a data generation device **100**, a controller **200**, a providing device **300**, an external communication network **400**, and sensors. The data generation device **100**, the controller **200**, and the sensors are connected to each other in a manner that allows communication between them via the external communication network **400**. The controller **200** and the providing device **300** are connected to each other in a manner that allows communication between them via the external communication network **400**.

[0028] The data generation device **100** includes first processing circuitry **101**, a first storage device **102**, and a first communication device **103**. The first storage device **102** stores a data generation program. The first processing circuitry **101** executes the data generation program stored in the first storage device **102** to execute various processes. The first processing circuitry **101** includes one or more processors. The data generation device **100** is connected to the external communication network **400** via the first communication device **103**.

[0029] The controller **200** includes second processing circuitry **201**, a second storage device **202**, and a second communication device **203**. The second storage device **202** stores a program. The second processing circuitry **201** executes the program stored in the second storage device **202** to execute various processes. The second processing circuitry **201** includes one or more processors. The controller **200** is connected to the external communication network **400** via the second communication device **203**.

[0030] The providing device **300** includes third processing circuitry **301**, a third storage device **302**, and a third communication device **303**. The third storage device **302** stores a program. The third processing circuitry **301** executes the program stored in the third storage device **302** to execute various processes. The third processing circuitry **301** includes one or more processors. The providing device **300** is connected to the external communication network **400** via the third communication device **303**.

[0031] Each of the first processing circuitry **101**, the second processing circuitry **201**, and the third processing circuitry **301** may include hardware circuitry including one or more dedicated hardware circuits such as an application-specific integrated circuit (ASIC) that execute at least some of the various processes or a combination thereof. Alternatively, each of the first processing circuitry **101**, the second processing circuitry **201**, and the third processing circuitry **301** may include a combination of one or more processors and one or more dedicated hardware circuits. Each of the first storage device **102**, the second storage device **202**, and the third storage device **302** includes a memory such as a RAM and a ROM. The memory, or a computer-readable medium, includes any type of media that are accessible by general-purpose computers and dedicated computers.
[0032] The sensors of the traffic service providing system **10** will now be described.

Sensors of Traffic Service Providing System **10**

[0033] The sensors of the traffic service providing system **10** collect moving body information, which indicates the positions and behaviors of moving bodies **800** in the real world, at multiple points in time. The moving bodies **800** are objects that move in the real world. Examples of the moving body **800** include a vehicle **600**, a pedestrian **700**, a bicycle, and an animal. The vehicle **600** includes a motorcycle.
[0034] The vehicle **600** collects moving body information using a vehicle on-board sensor. That is, the vehicle **600** serves as a sensor. Examples of the vehicle on-board sensor of the vehicle **600** include a vehicle speed sensor, an accelerator sensor, a brake sensor, a steering sensor, and an acceleration sensor. The acceleration sensor is, for example, an inertial measurement unit (IMU).
[0035] The vehicle **600** further includes an external camera, a sonar, and a position information acquisition system as the vehicle on-board sensor. The external camera and the sonar mounted on the vehicle **600** collect information related to the distance between the vehicle **600** and one or more other objects located around the vehicle **600**, thereby generating an observational dataset. The vehicle **600** may include a light detection and ranging (LiDAR) sensor as a sensor that has the same features as those of the external camera and the sonar. Examples of the position information acquisition system include a global navigation satellite system (GNSS), a real-time kinematic (RTK)-enabled device, and a LiDAR sensor. The vehicle **600** sends the moving body information collected by the vehicle on-board sensors via the external communication network **400**.
[0036] The information processing terminal **500**, which is carried by the pedestrian **700**, collects the position information of the pedestrian **700**. That is, the information processing terminal **500** carried by the pedestrian **700** serves as a sensor. Examples of the information processing terminal **500** carried by the pedestrian **700** include a smartphone carried by the pedestrian **700**. Examples of the information processing terminal **500** further include a wearable device and a tablet device. Examples of the wearable terminal include a ring-type terminal worn on the wrist and a necklace-type terminal worn around the neck.
[0037] The roadside sensors **900** are installed on the road. The roadside sensors **900** include, for example, traffic signals **910**, roadside cameras **920**, and LiDAR sensors installed on the roadside.
[0038] Each traffic signal **910** provides information related to changes in the state of the traffic infrastructure, such as the time at which the traffic signal **910** turns green and the time during which the traffic signal **910** remains green, to the controller **200**.
[0039] Each roadside camera **920** collects moving body information around it. Examples of the roadside cameras **920** include visible light cameras and infrared cameras. The roadside camera **920** acquires observational datasets arranged in chronological order by continuously observing the moving body **800** at fixed time intervals. The LiDAR sensors installed on the road acquire point cloud datasets arranged in chronological order, by continuously observing the moving body **800** at fixed time intervals.

Features of Controller **200**

[0040] The controller **200** periodically acquires the moving body information related to multiple moving bodies **800** via the second communication device **203** from multiple information processing terminals **500**, multiple vehicles **600**, and multiple roadside sensors **900**. The points in time at which the controller **200** acquires the moving body information related to each moving body **800** may differ from one another.

[0041] The controller **200** collects moving body information, including, for example, vehicle information (e.g., a vehicle identification number (VIN) of the vehicle **600**) and information related to the vehicle **600** (e.g., a vehicle speed, a travel direction, and a travel route and a position).

[0042] The controller **200** stores the moving body information related to each moving body **800** in the second storage device **202**, associating with the point in time at which it was acquired. The second processing circuitry **201** of the controller **200** uses the moving body information stored in the second storage device **202** to calculate predicted moving body information, which indicates the position of each moving body **800** after the point in time at which the moving body information was acquired. The predicted moving body information re-creates the real-world traffic environment in a virtual space.

[0043] The controller **200** communicates with multiple providing devices **300** via the external communication network **400**. For example, the controller **200** generates a control signal for each providing device **300** based on the predicted moving body information. The controller **200** sends each control signal to the corresponding providing device **300**.

[0044] Upon receiving the control signal, the providing device **300** provides traffic services to the user **20** of the providing device **300** based on the control signal. That is, the traffic service providing system **10** provides traffic services to assist the user **20** of each of the providing devices **300** based on the predicted moving body information.

Example of Providing Device **300**

[0045] The vehicle **600** provides traffic services to the user **20** of the vehicle **600** based on the control signals sent from the controller **200**. That is, the vehicle **600** serves as the providing device **300**.

[0046] The vehicle **600** includes vehicle on-board processing circuitry, a braking system, a steering system, directional indicators, speakers, and displays. The displays of the vehicle **600** display traffic services for the user **20** of the vehicle **600**. For example, the vehicle on-board processing circuitry uses the predicted moving body information to display a vehicle approach notification or traffic information on the displays of the vehicle **600**. For example, the vehicle on-board processing circuitry uses the predicted moving body information to issue a vehicle approach alert to the user **20** of the vehicle **600** via the speakers of the vehicle **600**.

[0047] For example, the vehicle on-board processing circuitry uses the predicted moving body information to control the braking system of the vehicle **600**, thereby decelerating or stopping the vehicle **600**. For example, the vehicle on-board processing circuitry uses the predicted moving body information to control the steering system of the vehicle **600**, thereby controlling the steering of the vehicle **600**. The vehicle on-board processing circuitry may also control the directional indicators in addition to controlling the steering.

[0048] The information processing terminal **500** provides traffic services to the user **20** of the information processing terminal **500** based on the control signals sent from the controller **200**. That is, the information processing terminal **500** serves as the providing device **300**. Examples of the information processing terminal **500**, which serves as the providing device **300**, include a smartphone carried by the user **20** of the vehicle **600**. For example, the information processing terminal **500** uses the predicted moving body information to display a vehicle approach notification or traffic information on the display of the information processing terminal **500**.

Features of Data Generation Device **100**

[0049] Depending on the type of traffic service that the providing device **300** provides to the user **20**, predicted moving body information that is highly accurate may be required. For example, for the controller **200** to generate control signals that cause the vehicle on-board processing circuitry of the vehicle **600** to execute steering control, highly accurate predicted moving body information is required. Such information allows the controller **200** to accurately acquire the position and behavior of an object around the vehicle **600** in addition to the position and behavior of the vehicle **600**.

[0050] To improve the accuracy of the predicted moving body information, the controller **200** acquires a synchronized observational dataset from the data generation device **100**. Based on the moving body information stored in the second storage device **202** and the synchronized observational dataset, the controller **200** generates highly accurate predicted moving body information.

[0051] The data generation device **100** generates a synchronized observational dataset. The synchronized observational dataset is an observational dataset in which the points in time have been synchronized based on reference data. The reference data is selected from the data included in observational datasets based on accuracy information.

[0052] The first communication device **103** of the data generation device **100** obtains multiple observational datasets from multiple roadside cameras **920** via the external communication network **400**. The observational datasets are obtained by continuously observing, from different positions, two or more objects that have a changing distance between them. This indicates that multiple observational datasets are also obtained by observing the same moving body **800** from multiple viewpoints. Synchronizing the above-described observational datasets generates observational datasets in which the position and behavior of the moving body **800** were observed at the same point in time from multiple viewpoints.

[0053] If the data generation device **100** synchronizes observational datasets using low-accuracy reference data from which the position of an object subject to observation is not accurately identifiable, the accuracy of the position of the moving body **800** in the synchronized observational datasets would decrease. This would also lower the accuracy of the predicted moving body information generated based on the synchronized observational datasets. To solve this problem, the data generation device **100** executes a synchronization process, which will be described later, to generate a synchronized observational dataset to limit the decrease in the accuracy of the position of the moving body **800**.

Communication in Traffic Service Providing System **10**

[0054] FIG. **2** illustrates the communication executed by the vehicle **600**, which serves as the providing device **300**, the controller **200**, the data generation device **100**, and the roadside camera **920** when a synchronized observational dataset is required in the traffic service providing system **10** of the embodiment.

[0055] In the example of FIG. **2**, the user **20** of the vehicle **600** requests that the vehicle **600** provide traffic services. Upon being requested by the user **20** to provide traffic services, the vehicle **600** sends a request for control signals to the controller **200**.

[0056] The controller **200** determines whether a synchronized observational dataset is necessary to generate the requested control signals. When determining that a synchronized observational dataset is necessary, the controller **200** requests the synchronized observational dataset from the data generation device **100**.

[0057] Upon receiving a request for a synchronized observational dataset, the data generation device **100** requests observational datasets from multiple roadside cameras **920**. In this case, the data generation device **100** selects, from the roadside cameras **920**, those installed in areas where the acquisition of information beneficial for providing traffic services to the vehicle **600** is expected. The data generation device **100** requests observational datasets from the selected roadside cameras **920**. Upon receiving a request for observational datasets, each roadside camera **920** sends the observational datasets to the data generation device **100**. Based on accuracy information, the data generation device **100** selects reference data from the data contained in the received observational datasets. The data generation device **100** executes the synchronization process to synchronize the points in time of multiple observational datasets based on the reference data, thereby generating a synchronized observational dataset. The data generation device **100** sends the synchronized observational dataset to the controller **200**.

[0058] Upon receiving the synchronized observational dataset, the controller **200** generates predicted moving body information based on the moving body information stored in the second storage device **202** and the synchronized observational dataset. The controller **200** generates control signals based on the predicted moving body information. The controller **200** sends the control signals to the vehicle **600**.

[0059] The vehicle **600** provides traffic services to the user **20** based on the acquired control signals.

Processes Executed by Data Generation Device **100**

[0060] The flow of the synchronization process, in which the data generation device **100** generates a synchronized observational dataset, will now be described in detail with reference to FIGS. **3** to **10**.

[0061] FIG. **3** is a flowchart illustrating a series of processes executed by the data generation device **100** in the traffic service providing system **10**.

[0062] The first storage device **102** of the data generation device **100** stores a data generation program that instructs the first processing circuitry **101** to execute the series of processes. The data generation device **100** executes the series of processes shown in FIG. **3** in accordance with the data generation program stored in the first storage device **102**. The data generation device **100** executes the series of processes when receiving a request for a synchronized observational dataset from the controller **200** via the first communication device **103**.

[0063] As shown in FIG. **3**, upon initiating the series of processes, the data generation device **100** acquires multiple observational datasets, which have been obtained through continuous observation, over a predetermined period of time from multiple roadside cameras **920** via the first communication device **103** in step S**11**. Continuous observation refers to the act of carrying out observations continuously at fixed time intervals. In the process of step S**11**, the data generation device **100** selects, from the roadside cameras **920**, those installed in areas where the acquisition of information beneficial for providing traffic services to the vehicle **600** is expected. The data generation device **100** requests the observational datasets obtained through continuous observation from the selected roadside cameras **920**. Thus, upon receiving a request for the observational datasets, each roadside camera **920** starts to send video datasets to the data generation device **100**. The data generation device **100** acquires the observational datasets during a predetermined period of time after requesting the observational datasets.

[0064] The observational datasets acquired by the roadside cameras **920** will now be described in detail with reference to FIG. **4**.

[0065] FIG. **4** is a schematic diagram illustrating an example of the acquisition of the observational datasets from the roadside cameras **920**. Two or more objects that have a changing distance between them correspond to a first vehicle **601** and a second vehicle **602**. The roadside cameras **920** that continuously observe the first vehicle **601** and the second vehicle **602** from different positions refer to a first roadside camera **921**, a second roadside camera **922**, and a third roadside camera **923**. The first roadside camera **921**, the second roadside camera **922**, and the third roadside camera **923** are visible light cameras that capture color images at a predetermined frame rate. The first roadside camera **921**, the second roadside camera **922**, and the third roadside camera **923** each acquire a video dataset. The video dataset corresponds to an observational dataset obtained through continuous observation. The video dataset also records the information related to the point in time when the roadside camera **920** obtained the video dataset. The video dataset is a collection of still image datasets arranged in chronological order. Each still image dataset included in the video dataset also records a capture time.

[0066] The video dataset captured by the first roadside camera **921** is defined as the first video dataset **921**D. The video dataset captured by the second roadside camera **922** is defined as the second video dataset **922**D. The video dataset captured by the third roadside camera **923** is referred to as the third video dataset **923**D.

[0067] The first roadside camera **921** sends the first video dataset **921**D to the data generation device **100** via the external communication network **400**. The second roadside camera **922** sends the second video dataset **922**D to the data generation device **100** via the external communication network **400**. The third roadside camera **923** sends the third video dataset **923**D to the data generation device **100** via the external communication network **400**. When the data generation device **100** requests observational datasets and then a predetermined period of time elapses, the data generation device **100** advances the process to step S**14**.

[0068] In step S**14**, the data generation device **100** selects as reference data, from the first video dataset **921**D, the second video dataset **922**D, and the third video dataset **923**D, a still image dataset in which the first vehicle **601** and the second vehicle **602** are clearly captured. That is, the data generation device **100** selects the reference data from multiple observational datasets acquired via the first communication device **103** based on accuracy information.

Method for Selecting Reference Data

[0069] The method by which the data generation device **100** selects the reference data in step S**14** will now be described in detail with reference to FIGS. **5** and **6**.

[0070] In the embodiment, the still image dataset shown in FIG. **5** is a first still image dataset **921**D_FRA included in the first video dataset **921**D, which was captured by the first roadside camera **921**. The first still image dataset **921**D_FRA clearly shows the first vehicle **601** and the second vehicle **602** as depicted in FIG. **4**. The first still image dataset **921**D_FRA records the capture time at which the first roadside camera **921** obtained the first still image dataset **921**D_FRA.

[0071] The first processing circuitry **101** uses a known image recognition algorithm to execute a process on the still image dataset included in each of the first video dataset **921**D, the second video dataset **922**D, and the third video dataset **923**D. In this process, the first processing circuitry **101** detects an object included in each still image dataset and sets a detection frame for the detected object. The detection frame may be referred to as a bounding box. The detection frame is the smallest rectangular frame that contains the entire object subject to detection. For example, the first processing circuitry **101** may use a pre-trained machine learning model to set a detection frame for the object in a still image dataset. The first processing circuitry **101** may detect an object through template matching with various sample images and then set a detection frame surrounding the detected object. In the first still image dataset **921**D_FRA shown in FIG. **5**, each detection frame is indicated by a dotted line. The detection frames in the first still image dataset **921**D_FRA include a first detection frame **1**B and a second detection frame **2**B.

[0072] Next, the first processing circuitry **101** determines whether the still image dataset includes two or more objects that have a changing distance between them. For example, when two detection frames that have a changing distance between them exist between two chronologically-ordered still image datasets obtained through continuous observation, the first processing circuitry **101** identifies the objects respectively contained by the two detection frames as two objects that have a changing distance between them. The first detection frame **1**B and the second detection frame **2**B in the first still image dataset **921**D_FRA, as shown in FIG. **5**, correspond to the two detection frames that have a changing distance between them. That is, the first still image dataset **921**D_FRA includes two or more objects that have a changing distance between them.

[0073] Subsequently, the first processing circuitry **101** executes edge detection on objects within two or more detection frames that have a changing distance between them in the still image dataset. This enables the data generation device **100** to determine whether the contours of the object included within each detection frame of the still image dataset are clear. Whether the contours clear is one aspect of accuracy information.

[0074] In edge detection, the first processing circuitry **101** converts the target still image dataset into grayscale. The first processing circuitry **101** detects edges, which are areas where the gradient of pixel intensity changes rapidly, within the detection frame of the still image dataset converted to grayscale. For example, the first processing circuitry **101** calculates the gradient of pixel values for the pixels within the detection frame based on the differences in pixel values in the horizontal and vertical directions. When the gradient of the pixel value of a pixel is greater than or equal to a determination value, the first processing circuitry **101** determines the pixel as an edge. Examples of the pixel value include the brightness of a pixel. The first processing circuitry **101** may execute edge detection using known techniques, such as a Laplacian filter or the Canny method.

[0075] The still image data with a large number of detected edges within the detection frame shows clear contours of the object within the detection frame. The first processing circuitry **101** determines that the still image dataset is sharp when the edges detected in the object within the detection frame have an image that is greater than a predetermined threshold value. For example, the first processing circuitry **101** executes edge detection to determine that the first still image dataset **921**D_FRA shown in FIG. **5** is sharp.

[0076] The data generation device **100** identifies the type and position of each object within the detection frame and identifies the distances between the detected objects from a sharp still image dataset. The data generation device **100** executes this process to select reference data. The first processing circuitry **101** executes this process on the first still image dataset **921**D_FRA, as shown in FIG. **5**. Examples of the type of objects include vehicles, humans, animals, installations on roads, and buildings.

[0077] The first processing circuitry **101** uses the above-described image recognition algorithm to identify the objects contained within the first detection frame **1**B as the first vehicle **601**. Similarly, the first processing circuitry **101** uses the image recognition algorithm to identify the objects contained within the second detection frame **2**B as the second vehicle **602**. To identify objects within the detection frame, the first processing circuitry **101** uses position information of the first vehicle **601** and the second vehicle **602**, which the controller **200** has acquired, and the information indicating the position of the roadside cameras **920**. Based on these types of information, the first processing circuitry **101** calculates a reference point, on the still image dataset, of the objects contained within each detection frame. In the embodiment, the first processing circuitry **101** calculates the center of the objects contained within each detection frame as the reference point on the still image dataset.

[0078] FIG. **6** is a schematic diagram illustrating the positions of the first vehicle **601** and the second vehicle **602**, calculated by the data generation device **100** from the first still image dataset **921**D_FRA shown in FIG. **5**. As shown in FIG. **6**, the first processing circuitry **101** sets the center of the first vehicle **601** on the first still image dataset **921**D_FRA to a first center **1**BC. Similarly, the first processing circuitry **101** sets the center of the second vehicle **602** on the first still image dataset **921**D_FRA to a second center **2**BC.

[0079] The first storage device **102** of the data generation device **100** stores data representing the dimensions and center positions of the first vehicle **601** and the second vehicle **602**. The first processing circuitry **101** uses the data representing the dimensions and center position of the first vehicle **601**, which are stored in the first storage device **102**, and uses the position of the first center **1**BC on the first still image dataset **921**D_FRA to calculate a position **601***oe* of a first outer edge, which is the position of the outer edge of the first vehicle **601**. Similarly, the first processing circuitry **101** uses the data representing the dimensions and center position of the second vehicle **602**, which are stored in the first storage device **102**, and uses the position of the second center **2**BC on the first still image dataset **921**D_FRA to calculate a position **602***oe* of a second outer edge, which is the position of the outer edge of the second vehicle **602**. The first processing circuitry **101** calculates a first inter-vehicle distance **30_1**, which is set between the position **601***oe* of the first outer edge and the position **602***oe* of the second outer edge. As shown in FIG. **6**, the first inter-vehicle distance **30_1** is set between a front end **601***oe*F at the position of the first outer edge and a rear end **602***oe*R at the position of the second outer edge.

[0080] The first still image dataset **921**D_FRA clearly shows the first vehicle **601** and the second vehicle **602**. Therefore, the position of the first vehicle **601** in the real world matches the position **601***oe* of the first outer edge calculated by the first processing circuitry **101**. Similarly, the position of the second vehicle **602** in the real world matches the position **602***oe* of the second outer edge calculated by the first processing circuitry **101**. Accordingly, the first inter-vehicle distance **30_1** is equal to the distance between the first vehicle **601** and the second vehicle **602** in the real world. Thus, the first processing circuitry **101** determines that the first inter-vehicle distance **30_1** is the distance between the first vehicle **601** and the second vehicle **602** in the real world.

[0081] The data generation device **100** executes the process of step S**14** to select, as reference data, a still image dataset that clearly shows the type of objects, the distance between the objects, and the capture time of the still image dataset, from the still image dataset contained in each of the first video dataset **921**D, the second video dataset **922**D, and the third video dataset **923**D. In the embodiment, the first still image dataset **921**D_FRA corresponds to the reference data. Whether the type of an object is clear is one aspect of the accuracy information. Whether the distance between objects is clear is one aspect of the accuracy information. Whether the capture time is clear is one aspect of the accuracy information. That is, the first processing circuitry **101** of the data generation device **100** selects the reference data based on the accuracy information.

[0082] In the process of step S**15**, the data generation device **100** determines whether it has selected reference data from the data contained in the observational datasets acquired from multiple roadside cameras **920**. In the process of step S**15**, when the data generation device **100** fails to select reference data from the data included in the acquired observational datasets (step S**15**: NO), the data generation device **100** advances the process to step S**100**.

[0083] In step S**100**, the data generation device **100** skips the providing of the synchronized observational datasets to the controller **200**, thereby terminating the series of processes.

[0084] When the acquired observational datasets do not include a still image dataset that clearly shows the type of objects, the distance between the objects, and the capture time of the still image dataset, the data generation device **100** does not select reference data. For example, when the acquired observational datasets do not include a still image dataset that clearly shows the contours of the objects through edge processing, no reference data is selected. For example, FIG. **7** is an example of a still image dataset that does not clearly show the contours of the objects.

[0085] In the process of step S**15**, when the data generation device **100** selects reference data from the data included in the acquired observational datasets (step S**15**: YES), the data generation device **100** advances the process to step S**16**.

[0086] Method for Selecting Still Image Data from Video Data Corresponding to Same Moment as Reference Data

[0087] In the process of step S**16**, the first processing circuitry **101** selects, from the video datasets that do not include the still image dataset selected as the reference data, a still image dataset showing a situation where the distance between two or more objects is equal to that distance in the reference data.

[0088] In the embodiment, the first processing circuitry **101** selects, from each of the second video dataset **922**D and the third video dataset **923**D, a still image data showing a situation where the distance between two or more objects is equal to that distance in the reference data. The process executed by the first processing circuitry **101** on the second video dataset **922**D will now be described as an example.

[0089] In the process of step S**16**, the first processing circuitry **101** of the data generation device **100** analyzes multiple second still image datasets **922**D_FRA of the second video dataset **922**D. Thus, the first processing circuitry **101** determines whether the capture range of each second still image dataset **922**D_FRA includes objects identical to the ones recorded in the first still image dataset **921**D_FRA, which serves as the reference data. The method by which the first processing

circuitry **101** executes this determination will now be described with reference to FIGS. **7** and **8**.
[0090] FIG. **7** illustrates the second still image dataset **922**D_FRA. The second still image dataset **922**D_FRA records the capture time at which the second roadside camera **922** obtained the second still image dataset **922**D_FRA. The second still image dataset **922**D_FRA, shown in FIG. **7**, is blurrier than the first still image dataset **921**D_FRA, shown in FIG. **5**. That is, the contours of the first vehicle **601** and the second vehicle **602** captured in the second still image dataset **922**D_FRA are blurry.
[0091] Through the process of step S**14**, a detection frame is set for each still image dataset included in the second video dataset **922**D and the third video dataset **923**D, and the objects in the detection frame are identified.
[0092] FIG. **7** illustrates the first detection frame **1**B, which displays the first vehicle **601**, and the second detection frame **2**B, which displays the second vehicle **602**.
[0093] Based on the information described above, the first processing circuitry **101** calculates the center position of the object contained within each detection frame as the reference point on the still image dataset, from the position of the detection frame in the still image data. In this case, the first processing circuitry **101** also uses the data representing the dimensions and center position of the object, which are stored in the first storage device **102**. Further, the first processing circuitry **101** uses the position information of the object and the position information of the roadside camera **920**, which are acquired by the controller **200**. Thus, the first processing circuitry **101** calculates the center position, on the still image data, of the object contained within each detection frame in the still image data to align with the above-described information.
[0094] The first processing circuitry **101** selects a still image data for each video dataset where the distance between two or more objects is equal to that distance in the reference data.
[0095] FIG. **8** is a schematic diagram illustrating the positions of the first vehicle **601** and the second vehicle **602**, calculated by the data generation device **100** from the second still image dataset **922**D_FRA shown in FIG. **7**. As shown in FIG. **8**, the first processing circuitry **101** sets the center of the first vehicle **601** on the second still image dataset **922**D_FRA to the first center **1**BC. Similarly, the first processing circuitry **101** sets the center of the second vehicle **602** on the second still image dataset **922**D_FRA to the second center **2**BC.
[0096] The first processing circuitry **101** uses the data representing the dimensions and the center position of the first vehicle **601**, which are stored in the first storage device **102**, and uses the position of the first center **1**BC on the second still image dataset **922**D_FRA to calculate the position **601***oe* of the first outer edge, which is the position of the outer edge of the first vehicle **601**. Similarly, the first processing circuitry **101** uses the data representing the dimensions and the center position of the second vehicle **602**, which are stored in the first storage device **102**, and uses the position of the second center **2**BC on the second still image dataset **922**D_FRA to calculate the position **602***oe* of the second outer edge, which is the position of the outer edge of the second vehicle **602**. The first processing circuitry **101** calculates a second inter-vehicle distance **30_2**, which is set between the position **601***oe* of the first outer edge and the position **602***oe* of the second outer edge in the second still image dataset **922**D_FRA. As shown in FIG. **8**, the second inter-vehicle distance **30_2** in the second still image data **922**D_FRA is set between the front end **601***oe*F at the position of the first outer edge and the rear end **602***oe*R at the position of the second outer edge.
[0097] In the embodiment, the first inter-vehicle distance **30_1** in the first still image dataset **921**D_FRA matches the second inter-vehicle distance **30_2** in the second still image dataset **922**D_FRA. That is, the second still image dataset **922**D_FRA shows a situation where the distance between two objects is equal to that distance in the first still image dataset **921**D_FRA, which is the reference data.
[0098] From the data included in each observational dataset, the data generation device **100** selects the still image dataset that shows the situation equivalent to that of the reference data. Then, the

data generation device **100** advances the process to step S**18**.

Generation of Synchronized Observational Dataset

[0099] In step S**18**, the first processing circuitry **101** of the data generation device **100** calculates the magnitude of the deviation between the capture time of the reference data and the capture time of the still image dataset that indicates the same situation as that of the reference data in each observational dataset.

[0100] FIG. **9** is a schematic diagram illustrating the relationship between the video datasets before the synchronization process has been executed. In FIG. **9**, the video datasets are arranged on the time axis according to the capture times recorded in the still image datasets that are included in each video dataset. The start time of the first video dataset **921**D based on the capture time recorded in each still image dataset included in the first video dataset **921**D is A**1**. The first video dataset **921**D includes the first still image dataset **921**D_FRA, which serves as the reference data. The capture time recorded in the first still image dataset **921**D_FRA, which serves as the reference data, is A**2**.

[0101] The start time of the second video dataset **922**D based on the capture time recorded in each still image dataset included in the second video dataset **922**D is B**1**. The second video dataset **922**D includes the second still image dataset **922**D_FRA, which shows the same situation as that of the first still image dataset **921**D_FRA, which serves as the reference data. The capture time recorded in the second still image dataset **922**D_FRA is B**2**. The magnitude of the deviation between the capture time recorded in the first still image dataset **921**D_FRA and the capture time recorded in the second still image dataset **922**D_FRA is ΔT**2**.

[0102] The start time of the third video dataset **923**D based on the capture time recorded in each still image dataset included in the third video dataset **923**D is C**1**. The third video dataset **923**D includes the third still image dataset **923**D_FRA, which shows the same situation as the first still image dataset **921**D_FRA, which serves as the reference data. The capture time recorded in the third still image dataset **923**D_FRA is C**2**. The magnitude of the deviation between the capture time recorded in the first still image dataset **921**D_FRA and the capture time recorded in the third still image dataset **923**D_FRA is ΔT**3**.

[0103] The first processing circuitry **101** calculates, for each video dataset, the magnitude of the deviation between the capture time recorded in the reference data and the capture time recorded in the still image dataset that indicates a situation equivalent to that of the reference data. Then, the first processing circuitry **101** advances the process to step S**19**.

[0104] In step S**19**, the data generation device **100** uses the magnitude of the deviation between the above-described capture times calculated for the video datasets in step S**18** to offset the start time of the video dataset. As a result, the data generation device **100** generates a synchronized observational dataset by synchronizing the video datasets with a video dataset that includes reference data. Specifically, the data generation device **100** synchronizes observational datasets such that the still image datasets illustrating the situation in which the distance between two or more objects is equal to that in the reference data are treated as a still image dataset at the same point in time. In this case, these still image datasets are included in each of the observational datasets.

[0105] FIG. **10** is a schematic diagram illustrating the synchronized observational dataset. As shown in FIG. **9**, the magnitude of the deviations between the capture times of the first still image dataset **921**D_FRA, which serves as the reference data, and the second still image dataset **922**D_FRA, which shows a situation equivalent to that of the reference data in the second video dataset **922**D, is ΔT**2**. As shown in FIG. **10**, the first processing circuitry **101** offsets the start time of the second video dataset **922**D by a period corresponding to ΔT**2** earlier. Specifically, the first processing circuitry **101** generates a synchronized second video dataset **922**Dsyn by updating, to a value offset by the amount of time that corresponds to ΔT**2**, the capture time recorded in the still image dataset contained in the second video dataset **922**D. As a result, the capture time recorded in the second still image dataset **922**D_FRA, which is contained in the synchronized second video dataset **922**Dsyn, is A**2**, which is the same as the capture time of the first still image dataset **921**D_FRA. The start time of the synchronized second video dataset **922**Dsyn is B**1**−ΔT**2**.

[0106] Similarly, the magnitude of the deviations between the capture times of the first still image dataset **921**D_FRA, which serves as the reference data, and the third still image dataset **923**D_FRA, which shows a situation equivalent to that of the reference data in the third still image dataset **923**D, is ΔT**3**. As shown in FIG. **10**, the first processing circuitry **101** offsets the start time of the third video dataset **923**D by a period corresponding to ΔT**3** later. Specifically, the first processing circuitry **101** generates a synchronized third video dataset **923**Dsyn by updating, to a value offset by the amount of time that corresponds to ΔT**3**, the capture time recorded in the still image dataset contained in the third video dataset **923**D. As a result, the capture time recorded in the third still image dataset **923**D_FRA, which is contained in the synchronized third video dataset **923**Dsyn, is A**2**, which is the same as the capture time of the first still image dataset **921**D_FRA. The start time of the synchronized third video dataset **923**Dsyn is C**1**+ΔT**3**.

[0107] In this manner, the data generation device **100** synchronizes multiple observational datasets such that the data indicating the same situation as the reference data is treated as data at the same point in time, thereby generating a synchronized observational dataset.

[0108] The first processing circuitry **101** generates the synchronized second video dataset **922**Dsyn and the synchronized third video dataset **923**Dsyn, and then advances the process to step S**20**.

[0109] In the process of step S**20**, the data generation device **100** sends the first video dataset **921**D, the synchronized second video dataset **922**Dsyn, and the synchronized third video dataset **923**Dsyn to the controller **200** via the first communication device **103**.

[0110] After executing the process of step S**20**, the data generation device **100** terminates the series of processes.

Providing Traffic Services Using Synchronized Observational Dataset

[0111] The second processing circuitry **201** of the controller **200** receives multiple synchronized observational datasets from the data generation device **100** via the second communication device **203**. The controller **200** generates predicted moving body information using the moving body information stored in the second storage device **202** and using the synchronized observational datasets received from the data generation device **100**. The controller **200** generates control signals based on the generated predicted moving body information. The controller **200** sends the generated control signals to the providing device **300**.

[0112] Based on the control signals received from the controller **200** via the third communication device **303**, the third processing circuitry **301** of the providing device **300** provides traffic services to the user **20** of the providing device **300**.

Operation of Present Embodiment

[0113] The data generation device **100** uses the first still image dataset **921**D_FRA, which clearly shows the distance between the first vehicle **601** and the second vehicle **602**, as reference data. The data generation device **100** synchronizes the second video dataset **922**D such that the second still image data **922**D_FRA, which indicates the moment at which the distance between the first vehicle **601** and the second vehicle **602** is equal to that in the reference data, is treated as data at the same point in time as the reference data. Similarly, the data generation device **100** synchronizes the third video dataset **923**D based on the reference data. That is, the data generation device **100** synchronizes multiple observational datasets based on a highly accurate reference observational dataset.

Advantages of Present Embodiment

[0114] (1) The data generation device **100** synchronizes observational datasets while limiting a decrease in the accuracy in each observational dataset.

[0115] (2) If an object is clearly captured in a still image dataset, the contours of the object are identifiable. As a result, the distance between two or more objects is clear. The first processing circuitry **101** of the data generation device **100** selects the first still image dataset **921**D_FRA, in which the object is clearly captured, as reference data. This allows the data generation device **100** to synchronize observational datasets using, as the reference data, the still image dataset that clearly shows the distance between the first vehicle **601** and the second vehicle **602**.

[0116] (3) The data generation device **100** includes the first storage device **102**. The first storage device **102** stores data representing the center position and dimensions of each of the objects. The center of each object serves as a reference point. Based on the stored data, the first processing circuitry **101** of the data generation device **100** identifies even a blurry object in the still image data **922**D_FRA through image recognition processing and identifies the center of the object on the still image data **922**D_FRA. The first processing circuitry **101** determines the position of the outer edge of the object with reference to the data representing the center position and dimensions of the object and with reference to the center of the object on the still image dataset identified as the reference point. The first processing circuitry **101** calculates the distance between two objects based on the information related to the position of the outer edge. By executing such processes, even if the objects captured in a still image data are blurry, the first processing circuitry **101** of the data generation device **100** determines the distance between two objects captured in the still image dataset. Thus, when synchronizing the reference data with observational datasets other than the reference data, the first processing circuitry **101** more easily determines the observational dataset that should be treated as the data at the same point in time as the reference data.

[0117] (4) The greater the deviation between the time at which an observational dataset was obtained and the time in the predicted moving body information calculated using the obtained observational dataset, the older the data used for calculating the predicted moving body information. Thus, the greater the deviation between the time at which an observational dataset was obtained and the time in the predicted moving body information calculated using the obtained observational dataset, the more uncertain the calculated predicted moving body information. When the first processing circuitry **101** of the data generation device **100** fails to select reference data from the data contained in the observational data acquired from multiple roadside cameras **920** over a predetermined period of time (step S**15**: NO), it does not generate a synchronized observational dataset. As a result, the data generation device **100** skips the generation of synchronized observational datasets that include outdated information that may make the predicted moving body information uncertain when used for calculation.

[0118] (5) The data generation method executed by the data generation device **100** includes the step (step S**11**) in which the data generation device **100** obtains observational datasets from the sensors via the first communication device **103** over the predetermined period of time. In this case, the datasets are obtained by continuously observing two or more objects that have a changing distance between them from different positions. The data generation method executed by the data generation device **100** includes the step (step S**14**) in which the first processing circuitry **101** selects, as reference data, data that clearly shows the distance between two or more objects from the observational datasets. The data generation method executed by the data generation device **100** includes the step (step S**19**) in which the first processing circuitry **101** uses the reference data to synchronize the observational datasets to generate a synchronized observational dataset. In this case, the synchronized observational datasets each include data indicating a situation in which the distance between two or more objects is equal to that distance in the reference data, and the data is treated as data obtained at the same point in time as that of the reference data. The generated synchronized observational dataset is used to calculate the predicted moving body information corresponding to a point in time after the observational datasets were acquired.

[0119] By executing such a data generation method, the first processing circuitry **101** of the data generation device **100** refers to the data that clearly shows the distance between two or more objects to synchronize observational datasets such that the data in which the moment when the distance between the two or more objects is equal is treated as the data obtained at the same point

in time. That is, the data generation device **100** synchronizes observational datasets with reference to a highly accurate reference observational dataset. The data generation method allows the data generation device **100** to synchronize observational datasets while limiting a decrease in the accuracy in each observational dataset.
[0120] (6) The first storage device **102** of the data generation device **100** stores a data generation program that instructs the first processing circuitry **101** to execute processes. The data generation program causes the first processing circuitry **101** of the data generation device **100** to execute obtaining observational datasets from the sensors via the first communication device **103** over the predetermined period of time. In this case, the datasets are obtained by continuously observing two or more objects that have a changing distance between them from different positions. The data generation program causes the first processing circuitry **101** to select, as reference data, the data that clearly shows the distance between two or more objects from multiple observational datasets. The data generation program causes the first processing circuitry **101** to use the reference data to synchronize observational datasets, thereby generating a synchronized observational dataset. In this case, the synchronized observational datasets each include data indicating a situation in which the distance between two or more objects is equal to that distance in the reference data, and the data is treated as data obtained at the same point in time as that of the reference data. The generated synchronized observational dataset is used to calculate the predicted moving body information corresponding to a point in time after the observational datasets were acquired. In other words, the data generation program causes the first processing circuitry **101** of the data generation device **100** to synchronize multiple observational datasets with reference to the data that clearly shows the distance between two or more objects such that the data in which the moment when the distance between the two or more objects is equal is treated as the data obtained at the same point in time. That is, the data generation program causes the data generation device **100** to synchronize the synchronization process with reference to a highly accurate reference observational dataset. The data generation program allows the data generation device **100** to synchronize observational datasets while limiting a decrease in the accuracy in each observational dataset.
[0121] (7) The traffic service providing system **10** provides traffic services to the user **20** of the providing device **300** based on the predicted moving body information generated using the synchronized observational dataset generated by the data generation device **100**. The synchronized observational dataset generated by the data generation device **100** is synchronized with reference data that clearly shows the distance between two or more objects, and is thus highly accurate. This increases the accuracy of the predicted moving body information generated by the controller **200** based on the highly accurate synchronized observational dataset. Thus, the traffic service providing system **10** provides the user **20** of the providing device **300** with traffic services that are based on high-quality predicted moving body information generated using the highly accurate synchronized observational dataset.
Modifications
[0122] The present embodiment may be modified as follows. The present embodiment and the following modifications can be combined as long as they remain technically consistent with each other.
[0123] The data acquired by the data generation device **100** as an observational dataset obtained through continuous observation is not limited to a video dataset. For example, the data generation device **100** may acquire a collection of point cloud datasets obtained from a LiDAR sensor and arranged in chronological order, as an observational dataset obtained through continuous observation.
[0124] In observational datasets acquired by the first communication device **103** of the data generation device **100**, at least one of the data type, image resolution, observation period, and observation duration may differ for each observational dataset.
[0125] The first processing circuitry **101** of the data generation device **100** may synchronize

different types of observational datasets. For example, the first processing circuitry **101** may synchronize a video dataset acquired from a camera sensor with a collection of point cloud datasets acquired from a LiDAR sensor and arranged in chronological order.

[0126] When selecting reference data, the data generation device **100** may select the reference data from the data that is included in the observational datasets obtained from sensors which satisfy a predetermined condition. For example, the data generation device **100** may select one still image dataset as the reference data from the still image datasets included in the video dataset acquired from a roadside camera **920** with an image resolution that is greater than or equal to a predetermined threshold value. The image resolution is one aspect of accuracy information. For example, the data generation device **100** may select one point cloud dataset as reference data from those included in a collection of point cloud datasets that are arranged in chronological order and acquired from a LiDAR sensor with the resolution of observation accuracy that is greater than a predetermined standard. The resolution of observation accuracy is one aspect of accuracy information.

[0127] The data generation device **100** may acquire an observational dataset from a sensor mounted on a moving body **800**. The data generation device **100** may acquire information indicating the status of the moving body **800** equipped with the sensor when acquiring an observational dataset from the sensor mounted on the moving body **800**. The information indicating the status of the moving body **800** includes, for example, the speed of the moving body **800** and the position of the moving body **800**. For example, the data generation device **100** may obtain a video dataset from an external camera mounted on the vehicle **600**. The data generation device **100** may obtain information related to at least one of the speed and position of the vehicle **600**, in addition to the video dataset, from the vehicle **600**.

[0128] The accuracy information used by the first processing circuitry **101** of the data generation device **100** to select reference data may be whether the front end of a first object and the edge of a second object located in front of the first object in the direction facing the first object are recognizable. For example, the first processing circuitry **101** may select reference data on the condition that a first front end **601**P of the first vehicle **601** and a second rear end **602**P of the second vehicle **602**, as shown in FIG. **4**, are recognizable. The distance between the first front end **601**P and the second rear end **602**P is equal to the distance between the first vehicle **601** and the second vehicle **602**. Thus, even if the first processing circuitry **101** selects the reference data based on the above-described condition, the data generation device **100** provides advantages equivalent to those of the embodiment.

[0129] The reference point is not limited to the center of the object contained within a detection frame. For example, the first front end **601**P and the second rear end **602**P, as shown in FIG. **4**, are reference points in a modification.

[0130] The accuracy information used by the first processing circuitry **101** of the data generation device **100** to select reference data may be whether characters written on an object are readable. For example, the first processing circuitry **101** may select reference data on the condition that the characters written on the object captured in a still image data are readable. When the characters written on the object captured in the still image data are readable, the still image data is clear. Thus, even if the first processing circuitry **101** sets the above-described condition as a condition for selecting reference data, the data generation device **100** provides advantages equivalent to those of the embodiment.

[0131] The accuracy information used by the first processing circuitry **101** of the data generation device **100** to select reference data may be whether the S/N ratio of a still image dataset or a point cloud dataset is greater than a predetermined value. For example, the first processing circuitry **101** may select the reference data on the condition that the S/N ratio of a still image data or point cloud dataset is greater than or equal to the predetermined value. The still image data or point cloud dataset with a S/N greater than or equal to the predetermined value is clear data with relatively little noise. Thus, even if the first processing circuitry **101** sets the above-described condition as a condition for selecting reference data, the data generation device **100** provides advantages equivalent to those of the embodiment.

[0132] In the embodiment, the data generation device **100** generates synchronized observational datasets by offsetting the start time for each observational dataset. The method for synchronizing observational datasets is not limited to the one described in the embodiment. For example, the data generation device **100** may add, to an observational data, offset amount data that is necessary to synchronize the observational dataset, and then send the synchronized observational dataset to the controller **200**.

[0133] Instead, for example, the data generation device **100** may generate a new synchronized observational dataset based on reference data and multiple synchronized observational datasets. That is, the data generation device **100** may generate a synchronized observational dataset that includes multiple synchronized observational datasets obtained from different positions.

[0134] The data generation device **100** and the controller **200** of the traffic service providing system **10** may be a single device. That is, the data generation device **100** may have the features of the controller **200** in the embodiment. In this case, the traffic service providing system **10** includes multiple providing devices **300**, a data generation device **100**, and multiple roadside sensors **900**.

[0135] The services offered by the traffic service providing system **10** to the user **20** via the vehicle **600** may be employed in various advanced safety technologies. Examples of advanced safety technologies include pre-crash safety (PCS), adaptive cruise control (ACC), lane keeping assist (LKA), and lane change assist (LCA).

[0136] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A data generation device, comprising: a communication device; and processing circuitry, wherein the communication device is configured to obtain, from sensors, observational datasets that have been obtained by continuously observing an object from different positions, the processing circuitry is configured to select reference data based on accuracy information related to the observational datasets, and generate a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data, and the synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

2. The data generation device according to claim 1, wherein the communication device is configured to obtain, as the observational datasets, video datasets obtained by capturing two or more objects from different positions, wherein a distance between the two or more objects is changing, and the processing circuitry is configured to select, as the reference data, from still image datasets included in each of the video datasets, a still image dataset in which each of the objects has a resolution that is greater than or equal to a threshold value.

3. The data generation device according to claim 2, wherein the processing circuitry is configured to calculate a distance between two of the objects based on a reference point, the reference point being set for each of the objects.

4. The data generation device according to claim 3, further comprising a storage device configured to store data representing dimensions of the objects categorized by type and data representing a position of the reference point set for each of the objects, and the processing circuitry is configured to: detect the object captured in the obtained still image dataset; identify a type of the detected object through image recognition processing; identify the reference point of the object on the still image dataset with reference to the data representing the dimensions of the objects categorized by type and the data representing the position of the reference point set for each of the objects, wherein the data representing the dimensions and the data representing the position are stored in the storage device; calculate a position of an outer edge of the detected object with reference to the identified reference point of the object on the still image dataset and with reference to the data representing the dimensions of the object and the data representing the position of the reference point; and calculate a distance between two different ones of the objects based on information related to the calculated position of the outer edge.

5. The data generation device according to claim 2, wherein the processing circuitry is configured to skip generation of the synchronized observational dataset when failing to acquire the observational dataset containing the reference data within a predetermined period of time.

6. A data generation method, comprising: obtaining, from sensors, observational datasets that have been obtained by continuously observing an object from different positions; selecting reference data based on accuracy information related to the observational datasets; and generating a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data, wherein the synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

7. A non-transitory storage medium storing a data generation program, wherein the data generation program is configured to cause a communication device to execute obtaining, from sensors, observational datasets that have been obtained by continuously observing an object from different positions, the data generation program is configured to cause processing circuitry to execute: selecting reference data based on accuracy information related to the observational datasets; and generating a synchronized observational dataset by synchronizing the observational datasets with each other based on the reference data, and the synchronized observational dataset is configured to be used to calculate a position of the object corresponding to a point in time after the observational datasets were obtained.

8. A traffic service providing system, comprising: the data generation device according to claim 1; a controller; and providing devices, wherein the data generation device is configured to: obtain the observational datasets from the sensors; generate the synchronized observational dataset by synchronizing the observational datasets with each other; and send the synchronized observational dataset to the controller, the controller is configured to generate predicted moving body information using the received synchronized observational dataset, and the controller and the providing devices are configured to communicate with each other, thereby providing a traffic service that assists a user of each of the providing devices based on the predicted moving body information.